Import Library

```
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader
from torchvision import datasets, models, transforms
from torchvision.models import EfficientNet_B0_Weights
import os
import time
import copy
import matplotlib.pyplot as plt
import random
from PIL import Image
import io

# Record the start time
start_time = time.time()
```

Mount Google Drive

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

Konfigurasi

```
FRAME_DIR = "/content/drive/MyDrive/datasetbaru/frame"
IMG_SIZE = (224, 224) # Ukuran input EfficientNetB0
BATCH_SIZE = 32
EPOCHS_INITIAL = 20
EPOCHS_FINE_TUNE = 30
INITIAL_LR = 1e-3
FINE_TUNE_LR = 1e-5

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Menggunakan device: {device}")
```

```
Menggunakan device: cuda
```

Setup Dataset (Data Augmentasi & Face Recognition)

```
def apply_jpeg_compression(img):

    # 1. Buat buffer di memori (RAM)
    buffer = io.BytesIO()

    # 2. Simpan gambar ke buffer sebagai JPEG dengan kualitas acak
    kualitas = random.randint(60, 80)
    img.save(buffer, "JPEG", quality=kualitas)

    # 3. Pindahkan "pointer" buffer kembali ke awal
    buffer.seek(0)

    # 4. Buka kembali gambar yang sudah terkompresi dari buffer
    img_compressed = Image.open(buffer).convert("RGB")

    # 5. Tutup buffer
    buffer.close()

    return img_compressed

data_transforms = {
    'train': transforms.Compose([
        # Ganti RandomResizedCrop dengan Resize + RandomCrop agar skala wajah tetap terjaga
        transforms.Resize(256),
        transforms.RandomCrop(IMG_SIZE), # atau RandomResizedCrop dengan scale=(0.8, 1.0)
        transforms.RandomHorizontalFlip(),

        # 2. Augmentasi "penyiksaan" (bekerja pada PIL Image)
        transforms.ColorJitter(brightness=0.3, contrast=0.3, saturation=0.2),

        transforms.RandomApply([
            transforms.GaussianBlur(kernel_size=(3, 7))
        ], p=0.3),

        # 3. Augmentasi Kompresi JPEG (SEKARANG SUDAH AMAN)
        #   Menggunakan fungsi helper kita
        transforms.RandomApply([
            transforms.Lambda(apply_jpeg_compression)
        ], p=0.4),

        # 4. Konversi ke Tensor (setelah semua augmentasi di atas)
        transforms.ToTensor(),

        # 5. Normalisasi (setelah ToTensor)
        transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
    ]),

    # 'validation' dan 'test' tetap sama (tidak perlu augmentasi)
    'validation': transforms.Compose([
        transforms.Resize(256),
        transforms.CenterCrop(IMG_SIZE),
        transforms.ToTensor(),
        transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
    ]),
    'test': transforms.Compose([
        transforms.Resize(256),
        transforms.CenterCrop(IMG_SIZE),
        transforms.ToTensor(),
        transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
    ]),
}
```

Klasifikasi Variabel & Dataloader

```
print("Memuat data wajah...")
image_datasets = {x: datasets.ImageFolder(os.path.join(FRAME_DIR, x), data_transforms[x]) for x in ['train', 'validation', 'test']}
dataloaders = {
    'train': DataLoader(image_datasets['train'], batch_size=BATCH_SIZE, shuffle=True, num_workers=2),
    'validation': DataLoader(image_datasets['validation'], batch_size=BATCH_SIZE, shuffle=True, num_workers=2),
    'test': DataLoader(image_datasets['test'], batch_size=BATCH_SIZE, shuffle=False, num_workers=2)
}
dataset_sizes = {x: len(image_datasets[x]) for x in ['train', 'validation', 'test']}
class_names = image_datasets['train'].classes
print(f"Kelas ditemukan: {class_names} -> 'asli':0, 'deepfake':1")
```

```
Memuat data wajah...
Kelas ditemukan: ['asli', 'deepfake'] -> 'asli':0, 'deepfake':1
```

Training Model

```
def train_model(model, criterion, optimizer, scheduler, num_epochs=25):
    since = time.time()
    best_model_wts = copy.deepcopy(model.state_dict())
    best_acc = 0.0

    # Lists to store history
    train_loss_history = []
    train_acc_history = []
    val_loss_history = []
    val_acc_history = []

    for epoch in range(num_epochs):
        print(f'Epoch {epoch+1}/{num_epochs}')
        print('-' * 10)
        for phase in ['train', 'validation']:
            if phase == 'train': model.train()
            else: model.eval()

            running_loss, running_corrects = 0.0, 0

            for inputs, labels in dataloaders[phase]:
                inputs, labels = inputs.to(device), labels.to(device)
                labels = labels.float().view(-1, 1)
                optimizer.zero_grad()
                with torch.set_grad_enabled(phase == 'train'):
                    outputs = model(inputs)
                    loss = criterion(outputs, labels)
                    if phase == 'train':
                        loss.backward()
                        optimizer.step()
                running_loss += loss.item() * inputs.size(0)
                preds = (torch.sigmoid(outputs) > 0.5).float()
                running_corrects += torch.sum(preds == labels.data)

            epoch_loss = running_loss / dataset_sizes[phase]
            epoch_acc = running_corrects.double() / dataset_sizes[phase]

            if phase == 'train':
                train_loss_history.append(epoch_loss)
                train_acc_history.append(epoch_acc.item())
            else:
                val_loss_history.append(epoch_loss)
                val_acc_history.append(epoch_acc.item())

            print(f'{phase} Loss: {epoch_loss:.4f} Acc: {epoch_acc:.4f}')

            if phase == 'validation' and epoch_acc > best_acc:
                best_acc = epoch_acc
                best_model_wts = copy.deepcopy(model.state_dict())
        if phase == 'train' and scheduler: scheduler.step()
        print()

    time_elapsed = time.time() - since
    print(f'Training complete in {time_elapsed // 60:.0f}m {time_elapsed % 60:.0f}s')
    print(f'Best val Acc: {best_acc:4f}')
    model.load_state_dict(best_model_wts)
    return model, train_loss_history, val_loss_history, train_acc_history, val_acc_history
```

Plot History

```
import matplotlib.pyplot as plt

def plot_history(train_loss, val_loss, train_acc, val_acc, title):
    epochs = range(len(train_loss))

    plt.figure(figsize=(12, 5))

    # Plot Loss
    plt.subplot(1, 2, 1)
    plt.plot(epochs, train_loss, label='Training Loss')
    plt.plot(epochs, val_loss, label='Validation Loss')
    plt.title(f'{title} - Loss')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.legend()
    plt.grid(True)

    # Plot Accuracy
    plt.subplot(1, 2, 2)
    plt.plot(epochs, train_acc, label='Training Accuracy')
    plt.plot(epochs, val_acc, label='Validation Accuracy')
    plt.title(f'{title} - Accuracy')
    plt.xlabel('Epochs')
    plt.ylabel('Accuracy')
    plt.legend()
    plt.grid(True)

    plt.tight_layout()
    plt.show()
```

Transfer Learning EfficientNet

```
weights = EfficientNet_B0_Weights.IMAGENET1K_V1
model = models.efficientnet_b0(weights=weights)

# Bekukan semua parameter base model
for param in model.features.parameters():
    param.requires_grad = False

# Ganti classifier
num_ftrs = model.classifier[1].in_features
model.classifier[1] = nn.Sequential(
        nn.Dropout(p=0.5), # Matikan 50% neuron secara acak saat training
        nn.Linear(num_ftrs, 1)
)

model = model.to(device)
criterion = nn.BCEWithLogitsLoss()
optimizer = optim.Adam(model.classifier.parameters(), lr=INITIAL_LR, weight_decay=1e-4)
optimizer_ft = optim.Adam(model.parameters(), lr=FINE_TUNE_LR, weight_decay=1e-4)
lr_scheduler = optim.lr_scheduler.StepLR(optimizer, step_size=7, gamma=0.1)

# Latih head
model, initial_train_loss, initial_val_loss, initial_train_acc, initial_val_acc = train_model(model, criterion, optimizer, lr_scheduler, num_epochs=EPOCHS_INITIAL)

# Plot history for initial training
plot_history(initial_train_loss, initial_val_loss, initial_train_acc, initial_val_acc, "Initial Training")
```

```
Downloading: "https://download.pytorch.org/models/efficientnet_b0_rwightman-7f5810bc.pth" to /root/.cache/torch/hub/checkpoints/efficientnet_b0_rwightman-7f5810bc.pth
100%|██████████| 20.5M/20.5M [00:00:00<00, 233MB/s]
Epoch 1/20
----------
train Loss: 0.6686 Acc: 0.5749
validation Loss: 0.6139 Acc: 0.7331

Epoch 2/20
----------
train Loss: 0.6168 Acc: 0.6739
validation Loss: 0.5778 Acc: 0.7211

Epoch 3/20
----------
train Loss: 0.6085 Acc: 0.6776
validation Loss: 0.5691 Acc: 0.7230

Epoch 4/20
----------
train Loss: 0.5846 Acc: 0.7038
validation Loss: 0.5489 Acc: 0.7517

Epoch 5/20
----------
train Loss: 0.5865 Acc: 0.6989
validation Loss: 0.5475 Acc: 0.7382

Epoch 6/20
----------
train Loss: 0.5821 Acc: 0.6980
validation Loss: 0.5440 Acc: 0.7382

Epoch 7/20
----------
train Loss: 0.5828 Acc: 0.6989
validation Loss: 0.5355 Acc: 0.7568

Epoch 8/20
----------
train Loss: 0.5759 Acc: 0.7022
validation Loss: 0.5422 Acc: 0.7297

Epoch 9/20
----------
train Loss: 0.5716 Acc: 0.7126
validation Loss: 0.5253 Acc: 0.7365

Epoch 10/20
----------
train Loss: 0.5450 Acc: 0.7284
validation Loss: 0.5383 Acc: 0.7331

Epoch 11/20
----------
train Loss: 0.5600 Acc: 0.7092
validation Loss: 0.5371 Acc: 0.7382

Epoch 12/20
----------
train Loss: 0.5484 Acc: 0.7242
validation Loss: 0.5324 Acc: 0.7517

Epoch 13/20
----------
train Loss: 0.5429 Acc: 0.7001
validation Loss: 0.5335 Acc: 0.7635

Epoch 14/20
----------
train Loss: 0.5681 Acc: 0.7121
validation Loss: 0.5327 Acc: 0.7551

Epoch 15/20
----------
train Loss: 0.5403 Acc: 0.7192
validation Loss: 0.5380 Acc: 0.7534

Epoch 16/20
----------
train Loss: 0.5537 Acc: 0.7209
validation Loss: 0.5410 Acc: 0.7348

Epoch 17/20
----------
train Loss: 0.5531 Acc: 0.7250
validation Loss: 0.5378 Acc: 0.7314

Epoch 18/20
----------
train Loss: 0.5421 Acc: 0.7134
validation Loss: 0.5445 Acc: 0.7280

Epoch 19/20
----------
train Loss: 0.5681 Acc: 0.7151
validation Loss: 0.5336 Acc: 0.7348

Epoch 20/20
----------
train Loss: 0.5587 Acc: 0.7213
validation Loss: 0.5410 Acc: 0.7331

Training complete in 16m 56s
Best val Acc: 0.763514
```

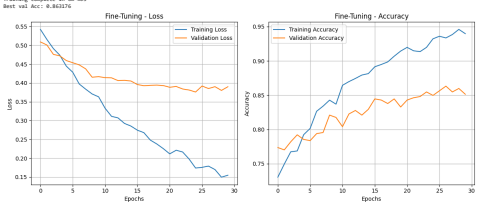
```
!pip install torchsummary

from torchsummary import summary

print("\n--- Model Summary ---")
# Pastikan model berada di device (CPU atau GPU) sebelum menampilkan summary
summary(model, input_size=(1, IMG_SIZE[0], IMG_SIZE[1]))
```

```
        BatchNorm2d-173          [-1, 672, 14, 14]           1,344
               SiLU-174          [-1, 672, 14, 14]               0
            Conv2d-175            [-1, 672, 7, 7]          16,800
        BatchNorm2d-176            [-1, 672, 7, 7]           1,344
               SiLU-177            [-1, 672, 7, 7]               0
AdaptiveAvgPool2d-178            [-1, 672, 1, 1]               0
            Conv2d-179            [-1, 28, 1, 1]          18,844
               SiLU-180            [-1, 28, 1, 1]               0
            Conv2d-181           [-1, 672, 1, 1]          19,488
           Sigmoid-182           [-1, 672, 1, 1]               0
SqueezeExcitation-183            [-1, 672, 7, 7]               0
            Conv2d-184            [-1, 192, 7, 7]         129,024
        BatchNorm2d-185            [-1, 192, 7, 7]             384
             MBConv-186            [-1, 192, 7, 7]               0
            Conv2d-187            [-1, 1152, 7, 7]        221,184
        BatchNorm2d-188            [-1, 1152, 7, 7]          2,304
               SiLU-189            [-1, 1152, 7, 7]               0
            Conv2d-190            [-1, 1152, 7, 7]          28,800
        BatchNorm2d-191            [-1, 1152, 7, 7]          2,304
               SiLU-192            [-1, 1152, 7, 7]               0
AdaptiveAvgPool2d-193            [-1, 1152, 1, 1]               0
            Conv2d-194            [-1, 48, 1, 1]          55,344
               SiLU-195            [-1, 48, 1, 1]               0
            Conv2d-196           [-1, 1152, 1, 1]          56,448
           Sigmoid-197           [-1, 1152, 1, 1]               0
SqueezeExcitation-198            [-1, 1152, 7, 7]               0
            Conv2d-199            [-1, 192, 7, 7]         221,184
        BatchNorm2d-200            [-1, 192, 7, 7]             384
 StochasticDepth-201            [-1, 192, 7, 7]               0
             MBConv-202            [-1, 192, 7, 7]               0
            Conv2d-203            [-1, 1152, 7, 7]        221,184
        BatchNorm2d-204            [-1, 1152, 7, 7]          2,304
               SiLU-205            [-1, 1152, 7, 7]               0
            Conv2d-206            [-1, 1152, 7, 7]          28,800
        BatchNorm2d-207            [-1, 1152, 7, 7]          2,304
               SiLU-208            [-1, 1152, 7, 7]               0
AdaptiveAvgPool2d-209            [-1, 1152, 1, 1]               0
            Conv2d-210            [-1, 48, 1, 1]          55,344
               SiLU-211            [-1, 48, 1, 1]               0
            Conv2d-212           [-1, 1152, 1, 1]          56,448
           Sigmoid-213           [-1, 1152, 1, 1]               0
SqueezeExcitation-214            [-1, 1152, 7, 7]               0
            Conv2d-215            [-1, 192, 7, 7]         221,184
        BatchNorm2d-216            [-1, 192, 7, 7]             384
 StochasticDepth-217            [-1, 192, 7, 7]               0
             MBConv-218            [-1, 192, 7, 7]               0
            Conv2d-219            [-1, 1152, 7, 7]        221,184
        BatchNorm2d-220            [-1, 1152, 7, 7]          2,304
               SiLU-221            [-1, 1152, 7, 7]               0
            Conv2d-222            [-1, 1152, 7, 7]          28,800
        BatchNorm2d-223            [-1, 1152, 7, 7]          2,304
               SiLU-224            [-1, 1152, 7, 7]               0
AdaptiveAvgPool2d-225            [-1, 1152, 1, 1]               0
            Conv2d-226            [-1, 48, 1, 1]          55,344
               SiLU-227            [-1, 48, 1, 1]               0
            Conv2d-228           [-1, 1152, 1, 1]          56,448
           Sigmoid-229           [-1, 1152, 1, 1]               0
SqueezeExcitation-230            [-1, 1152, 7, 7]               0
            Conv2d-231            [-1, 192, 7, 7]         221,184
```

Fine Tuning Model

```
print("\n--- Fine-Tuning Selesai Model ---")
for param in model.parameters():
    param.requires_grad = True

optimizer_ft = optim.Adam(model.parameters(), lr=FINE_TUNE_LR)
lr_scheduler_ft = optim.lr_scheduler.StepLR(optimizer_ft, step_size=7, gamma=0.1)

# Lanjutkan pelatihan
model, finetune_train_loss, finetune_val_loss, finetune_train_acc, finetune_val_acc = train_model(model, criterion, optimizer_ft, lr_scheduler_ft, num_epochs=EPOCHS_FINE_TUNE)

# Plot history for fine-tuning
plot_history(finetune_train_loss, finetune_val_loss, finetune_train_acc, finetune_val_acc, "Fine-Tuning")
```

```
--- Fine-Tuning Selesai ---
Epoch 1/30
----------
train Loss: 0.5422 Acc: 0.7304
validation Loss: 0.5089 Acc: 0.7736

Epoch 2/30
----------
train Loss: 0.5155 Acc: 0.7496
validation Loss: 0.5008 Acc: 0.7703

Epoch 3/30
----------
train Loss: 0.4913 Acc: 0.7675
validation Loss: 0.4759 Acc: 0.7821

Epoch 4/30
----------
train Loss: 0.4736 Acc: 0.7687
validation Loss: 0.4718 Acc: 0.7922

Epoch 5/30
----------
train Loss: 0.4646 Acc: 0.7924
validation Loss: 0.4594 Acc: 0.7855

Epoch 6/30
----------
train Loss: 0.4283 Acc: 0.8016
validation Loss: 0.4536 Acc: 0.7838

Epoch 7/30
----------
train Loss: 0.3972 Acc: 0.8265
validation Loss: 0.4479 Acc: 0.7939

Epoch 8/30
----------
train Loss: 0.3834 Acc: 0.8340
validation Loss: 0.4373 Acc: 0.7956

Epoch 9/30
----------
train Loss: 0.3706 Acc: 0.8428
validation Loss: 0.4152 Acc: 0.8209

Epoch 10/30
----------
train Loss: 0.3631 Acc: 0.8369
validation Loss: 0.4167 Acc: 0.8176

Epoch 11/30
----------
train Loss: 0.3328 Acc: 0.8644
validation Loss: 0.4140 Acc: 0.8041

Epoch 12/30
----------
train Loss: 0.3112 Acc: 0.8698
validation Loss: 0.4135 Acc: 0.8226

Epoch 13/30
----------
train Loss: 0.3071 Acc: 0.8744
validation Loss: 0.4061 Acc: 0.8277

Epoch 14/30
----------
train Loss: 0.2924 Acc: 0.8794
validation Loss: 0.4067 Acc: 0.8209

Epoch 15/30
----------
train Loss: 0.2853 Acc: 0.8814
validation Loss: 0.4051 Acc: 0.8294

Epoch 16/30
----------
train Loss: 0.2744 Acc: 0.8914
validation Loss: 0.3956 Acc: 0.8446

Epoch 17/30
----------
train Loss: 0.2679 Acc: 0.8948
validation Loss: 0.3927 Acc: 0.8429

Epoch 18/30
----------
train Loss: 0.2481 Acc: 0.8985
validation Loss: 0.3936 Acc: 0.8378

Epoch 19/30
----------
train Loss: 0.2378 Acc: 0.9068
validation Loss: 0.3942 Acc: 0.8446

Epoch 20/30
----------
train Loss: 0.2255 Acc: 0.9143
validation Loss: 0.3930 Acc: 0.8328

Epoch 21/30
----------
train Loss: 0.2115 Acc: 0.9197
validation Loss: 0.3884 Acc: 0.8429

Epoch 22/30
----------
train Loss: 0.2213 Acc: 0.9147
validation Loss: 0.3906 Acc: 0.8463

Epoch 23/30
----------
train Loss: 0.2165 Acc: 0.9135
validation Loss: 0.3839 Acc: 0.8480

Epoch 24/30
----------
train Loss: 0.1977 Acc: 0.9197
validation Loss: 0.3811 Acc: 0.8547

Epoch 25/30
----------
train Loss: 0.1738 Acc: 0.9322
validation Loss: 0.3758 Acc: 0.8497

Epoch 26/30
----------
train Loss: 0.1757 Acc: 0.9359
validation Loss: 0.3918 Acc: 0.8564

Epoch 27/30
----------
train Loss: 0.1789 Acc: 0.9334
validation Loss: 0.3851 Acc: 0.8632

Epoch 28/30
----------
train Loss: 0.1688 Acc: 0.9384
validation Loss: 0.3900 Acc: 0.8547

Epoch 29/30
----------
train Loss: 0.1495 Acc: 0.9459
validation Loss: 0.3881 Acc: 0.8598

Epoch 30/30
----------
train Loss: 0.1548 Acc: 0.9397
validation Loss: 0.3899 Acc: 0.8514

Training complete in 6m 40s
Best val Acc: 0.863176
```



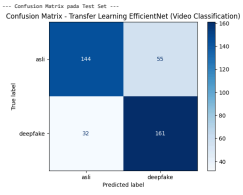
Evaluasi Model

```
print("\n--- EVALUASI AKHIR PADA TEST SET ---")
model.eval()
running_corrects = 0
with torch.no_grad():
    for inputs, labels in dataloaders['test']:
        inputs, labels = inputs.to(device), labels.to(device)
        labels = labels.float().view(-1, 1)
        outputs = model(inputs)
        preds = (torch.sigmoid(outputs) > 0.5).float()
        running_corrects += torch.sum(preds == labels.data)

accuracy = running_corrects.double() / dataset_sizes['test']
print(f'Akurasi akhir pada test set: {accuracy.item() * 100:.2f}%')
```

```
--- EVALUASI AKHIR PADA TEST SET ---
Akurasi akhir pada test set: 77.81%
```

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import numpy as np

print("\n--- Confusion Matrix pada Test Set ---")
model.eval()
all_labels = []
all_preds = []
with torch.no_grad():
    for inputs, labels in dataloaders['test']:
        inputs, labels = inputs.to(device), labels.to(device)
        outputs = model(inputs)
        preds = (torch.sigmoid(outputs) > 0.5).float()
        all_labels.extend(labels.data.cpu().numpy())
        all_preds.extend(preds.cpu().numpy())

cm = confusion_matrix(all_labels, all_preds)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=class_names)
disp.plot(cmap=plt.cm.Blues)
plt.title('Confusion Matrix - Transfer Learning EfficientNet (Video Classification)')
plt.show()
```

```
--- Confusion Matrix pada Test Set ---
```




```
from sklearn.metrics import precision_score, recall_score, f1_score, accuracy_score
import numpy as np

print("\n--- Metrik Evaluasi pada Test Set ---")
model.eval()
all_labels = []
all_preds = []
with torch.no_grad():
    for inputs, labels in dataloaders['test']:
        inputs, labels = inputs.to(device), labels.to(device)
        outputs = model(inputs)
        preds = (torch.sigmoid(outputs) > 0.5).float()
        all_labels.extend(labels.data.cpu().numpy())
        all_preds.extend(preds.cpu().numpy())

y_true = np.array(all_labels)
y_pred_labels = np.array(all_preds)

precision = precision_score(y_true, y_pred_labels)
recall = recall_score(y_true, y_pred_labels)
f1 = f1_score(y_true, y_pred_labels)
accuracy = accuracy_score(y_true, y_pred_labels)

print(f"Akurasi  : {accuracy:.4f}")
print(f"Presisi  : {precision:.4f}")
print(f"Recall   : {recall:.4f}")
print(f"F1-Score : {f1:.4f}")
```

```
--- Metrik Evaluasi pada Test Set ---
Akurasi  : 0.7781
Presisi  : 0.7454
Recall   : 0.8342
F1-Score : 0.7873
```

Simpan Model Terbaik

```
torch.save(model.state_dict(), 'deepfake_detector_efficientnet.pth')
print("\nModel telah disimpan.")

end_time = time.time()
total_time = end_time - start_time
print(f"\nTotal waktu eksekusi: {total_time // 60:.0f}m {total_time % 60:.0f}s")
```

```
Model telah disimpan.

Total waktu eksekusi: 31m 17s
```